ALBAR: ADVERSARIAL LEARNING APPROACH TO MITIGATE BIASES IN ACTION RECOGNITION

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ABSTRACT

Action recognition models frequently exploit spurious static cues such as background scenes and subject appearance, achieving high in-distribution accuracy while failing under distribution shift. We present ALBAR (Adversarial Learning approach to mitigate Biases in Action Recognition), a training framework that suppresses reliance on static spatial shortcuts without requiring bias attribute labels or auxiliary critic networks. ALBAR pairs each training video with a motionless counterpart constructed by tiling a single frame, then adversarially penalizes confident predictions on these static clips through negative cross-entropy and entropy maximization losses, complemented by a gradient penalty for training stability. This forces the model to ground its predictions in temporal motion features rather than static appearance. We also introduce an improved UCF101-SCUBA/SCUFO benchmark that replaces bounding-box masks with fine-grained segmentation masks obtained via SAMTrack, eliminating the background leakage present in prior evaluation protocols. On the HMDB51 benchmark, ALBAR achieves a contrasted accuracy of 53.22%, a 12.94 absolute point improvement over the previous state of the art (40.28%), corresponding to a 91.2% relative gain. On our improved UCF101 benchmark, ALBAR attains 26.23% contrasted accuracy versus 17.53% for the baseline, a 49.6% relative improvement, while preserving 94.98% IID accuracy. ALBAR further improves downstream performance on UCF-Crime anomaly detection (84.91 vs. 82.39 AUC) and THUMOS14 temporal action localization (55.20 vs. 54.89 mAP), demonstrating that motion-grounded representations transfer effectively to related video understanding tasks.

1 INTRODUCTION

Deep learning has driven remarkable progress in video action recognition, with modern architectures achieving near-human accuracy on standard benchmarks such as UCF101 (Soomro et al., 2012) and HMDB51 (Kuehne et al., 2011). However, a growing body of evidence reveals that these models often exploit spurious correlations—static cues like background scenes, object co-occurrences, and subject appearance—rather than learning the temporal motion patterns that truly define actions (Geirhos et al., 2020; Choi et al., 2019; Li et al., 2018). A model trained on videos of "swimming" may learn to recognize water rather than the swimming motion itself, leading to brittle predictions that collapse when the visual context changes. This shortcut learning phenomenon (Geirhos et al., 2020) severely limits the reliability and generalization of action recognition systems in real-world deployment.

Prior work has approached this problem from multiple angles. Scene debiasing methods (Choi et al., 2019; Ding et al., 2022) attempt to reduce background dependence, while data augmentation strategies such as StillMix (Li et al., 2023) and VideoMix (Yun et al., 2020) diversify training distributions. Background erasure techniques (Wang et al., 2021) physically remove scene information. However, many of these approaches require explicit bias attribute annotations, auxiliary discriminator networks, or are limited to addressing only one type of bias (e.g., background but not foreground appearance). Furthermore, existing evaluation protocols for measuring bias—particularly the UCF101-SCUBA/SCUFO benchmarks—rely on bounding-box masks for foreground extraction, which allows background information to leak into supposedly bias-controlled test sets, undermining the validity of evaluation results (Figure 1).

In this work, we make two complementary contributions that jointly advance both the methodology and evaluation of bias mitigation in action recognition. First, we propose **ALBAR** (Adversarial

Learning approach to mitigate Biases in Action Recognition), a training framework that suppresses reliance on static spatial cues without requiring bias attribute labels or auxiliary networks. The key insight is simple yet effective: for each training video, we construct a motionless counterpart by tiling a single frame across the temporal dimension and adversarially train the model to produce uninformative (maximum-entropy) predictions on these static inputs, while maintaining discriminative performance on the original motion clips (Figure 2). This is achieved through a combination of negative cross-entropy on static clips, entropy maximization, and a gradient penalty for training stability. ALBAR is architecture-agnostic and can be combined with existing augmentation-based debiasing methods for further gains.

Second, we introduce an **improved UCF101-SCUBA/SCUFO benchmark** that addresses a critical flaw in prior evaluation protocols. By replacing coarse bounding-box masks with fine-grained actor segmentation masks produced by SAMTrack (Cheng et al., 2023), our benchmark eliminates background leakage and provides a more rigorous assessment of whether models truly rely on motion versus static cues. The benchmark evaluates both background bias (SCUBA: foreground composited onto novel backgrounds from Places365 (Zhou et al., 2017), VQGAN-CLIP (Crowson et al., 2022), and synthetic textures) and foreground bias (SCUFO: static frame-stacked clips), along with a conflicting-foreground variant.

Our experiments demonstrate substantial improvements. On HMDB51, ALBAR achieves a contrasted accuracy of 53.22%, improving over the prior state-of-the-art StillMix (Li et al., 2023) by 12.94 absolute points (+91.2% relative). On our improved UCF101 benchmark, ALBAR reaches 26.23% contrasted accuracy compared to 17.53% for the baseline (+49.6% relative), while maintaining 94.98% IID accuracy (vs. 95.51% baseline). ALBAR also improves downstream task performance: UCF-Crime (Sultani et al., 2018) anomaly detection AUC increases from 82.39 to 84.91, and THUMOS14 (Jiang et al., 2014) temporal action localization mAP improves from 54.89 to 55.20. These results confirm that enforcing motion-grounded representations yields more robust and transferable video understanding.

Our contributions can be summarized as follows:

- We propose ALBAR, a label-free adversarial training framework that suppresses static-spatial shortcuts through entropy maximization on motionless clips and a gradient penalty, requiring no bias annotations or auxiliary networks.
- We introduce an improved UCF101-SCUBA/SCUFO benchmark using segmentation-based actor masks that eliminates background leakage, enabling more rigorous evaluation of bias in action recognition.
- We achieve state-of-the-art debiasing results on both HMDB51 and our improved UCF101 benchmark, with gains that transfer to downstream anomaly detection and temporal action localization tasks.

2 Related Work

Bias in Action Recognition. Action recognition models are known to exploit spurious correlations present in training data, a phenomenon broadly characterized as shortcut learning (Geirhos et al., 2020). Li et al. (2018) demonstrated that scene information alone can predict action labels with surprisingly high accuracy, revealing severe scene bias in datasets like UCF101 and HMDB51. Choi et al. (2019) further showed that models trained on these datasets heavily rely on background context, failing when actions are placed in unfamiliar scenes. Byvshev et al. (2022) analyzed how 3D convolutional architectures inherit and amplify such biases, while Weinzaepfel & Rogez (2021) introduced the Mimetics dataset to evaluate action recognition in out-of-context scenarios. More recently, Dave et al. (2024) provided a comprehensive analysis of shortcut dependencies in video models. These findings collectively motivate the development of debiasing methods that can force models to learn motion-based features rather than static contextual cues.

Debiasing Methods for Action Recognition. Several approaches have been proposed to mitigate bias in action recognition. Choi et al. (2019) introduced a scene-debiasing network (SDN) that adversarially removes scene information from learned representations. Ding et al. (2022) proposed FAME, which uses motion-aware feature extraction to reduce background dependence. Augmentation-based

methods have also shown promise: StillMix (Li et al., 2023) generates counterfactual training samples by mixing still frames with action videos, while VideoMix (Yun et al., 2020) and Mixup (Zhang et al., 2017) provide general video augmentation strategies. Wang et al. (2021) proposed background erasing to physically remove scene information during training. Duan et al. (2022) addressed representation bias through sample-level attention mechanisms. Zhai et al. (2023) proposed scene-debiasing for open-set action recognition using adversarial scene reconstruction. Unlike these methods, which often require explicit bias labels, auxiliary discriminator networks, or address only a single type of bias, ALBAR uses a simple adversarial objective on static clips that jointly mitigates both background and foreground appearance biases without additional annotations.

Benchmarks for Evaluating Bias. Evaluating the extent of bias in action recognition requires carefully designed benchmarks. Li et al. (2018) proposed Resound to quantify scene bias, while Li & Vasconcelos (2019) introduced REPAIR to measure and reduce representation bias in activity datasets. The SCUBA (Scene Change UCF101 Background Augmented) and SCUFO (Scene Change UCF101 Foreground Only) protocols (Li et al., 2023) evaluate models under controlled background and foreground manipulations. However, these protocols use bounding-box-based foreground extraction, which inevitably leaks background pixels into the composited test videos. Our improved benchmark replaces these coarse masks with fine-grained segmentation masks from SAMTrack (Cheng et al., 2023), built upon SAM (Kirillov et al., 2023), eliminating leakage and providing a more reliable evaluation. We source diverse backgrounds from Places365 (Zhou et al., 2017), VQGAN-CLIP (Crowson et al., 2022), and synthetic textures, and introduce a contrasted accuracy metric that jointly captures performance on both SCUBA and SCUFO conditions.

Adversarial Training for Fairness and Invariance. Adversarial training has been widely used to learn invariant representations across domains. Goodfellow et al. (2014) introduced generative adversarial networks, establishing the adversarial min-max framework. Xie et al. (2017) proposed controllable invariance through adversarial feature learning for domain adaptation. In the fairness literature, adversarial debiasing has been applied to remove sensitive attributes from learned representations (Elazar & Goldberg, 2018; Zhang et al., 2018). Bahng et al. (2020) proposed learning de-biased representations using intentionally biased auxiliary models. ALBAR draws inspiration from this line of work but takes a distinct approach: rather than training an auxiliary adversary to remove specific attributes, we adversarially train the *same* classifier to produce maximum-entropy predictions on motionless inputs, directly penalizing the model's ability to extract discriminative information from static cues. This eliminates the need for bias attribute labels and avoids the instability of training separate adversarial networks.

3 Benchmark Design

Evaluating robustness to static-cue shortcuts requires test distributions that isolate background and foreground biases. We introduce an improved evaluation protocol for UCF101 (Soomro et al., 2012) and HMDB51 (Kuehne et al., 2011) that eliminates background leakage present in prior work (Li et al., 2023; Choi et al., 2019) and provides fine-grained diagnostic metrics.

3.1 Overview

Our benchmark comprises three out-of-distribution (OOD) test conditions derived from standard action recognition datasets:

SCUBA (**S**tatic **Cu**es in the **B**ackground **A**ssessment) evaluates background bias. Original foreground actors are composited onto novel backgrounds drawn from external sources, so a model relying on scene context will fail.

SCUFO (**S**tatic **Cu**es in the **Fo**reground) evaluates foreground bias. A single frame from the SCUBA clip is tiled across all temporal positions, producing a clip with correct spatial appearance but no motion. A model exploiting static object or actor appearance without attending to dynamics will succeed on SCUFO, revealing reliance on foreground shortcuts.

Confl-FG (Conflicting Foreground) places a static foreground from one clip onto a novel background, creating a maximally adversarial condition in which both background and temporal cues are disrupted.

(a) Prior bounding-box protocol. Background leakage (e.g., snow) is visible around the actor.

(b) Our improved protocol with tight SAMTrack segmentation masks, eliminating leakage.

Figure 1: Comparison of foreground extraction protocols for SCUBA test clips. *(Left)*Prior work uses bounding-box masks that leak background pixels (snow visible in a skiing clip). *(Right)*Our segmentation-based masks tightly delineate the actor, preventing background information from contaminating the foreground composite.

3.2 Data Construction

Actor segmentation. Previous protocols (Li et al., 2023; Choi et al., 2019) extracted foreground regions using bounding-box masks, which inevitably leak background pixels into the foreground composite. Figure 1a illustrates this problem: snow remains visible around the actor in a skiing clip, allowing the model to exploit the leaked scene context. We replace bounding boxes with fine-grained per-frame actor segmentation masks produced by SAMTrack (Cheng et al., 2023), initialized from THUMOS-14 (Jiang et al., 2014) bounding-box annotations and refined via the Segment Anything Model (Kirillov et al., 2023). Every test video undergoes manual verification to ensure that the resulting masks tightly delineate the actor and that no background content leaks through (Figure 1b).

Background sources. To ensure that replacement backgrounds carry no action-relevant information, we draw from three diverse sources: (i) images from the Places365 test set (Zhou et al., 2017), providing naturalistic but action-unrelated scenes; (ii) images generated by VQGAN-CLIP (Crowson et al., 2022), offering stylistically varied synthetic backgrounds; and (iii) synthetic sinusoidal textures, which are devoid of any semantic content. This diversity prevents models from learning new spurious background–action correlations in the evaluation set.

Clip synthesis. Given a source video, the *SCUBA* clip is constructed by compositing the segmented foreground actor onto a randomly selected novel background for each frame. The *SCUFO* clip is then created by selecting a single frame from the SCUBA clip and tiling it across the full temporal extent, thereby preserving spatial appearance while removing all motion. The *Confl-FG* variant applies the same tiling procedure but places the static foreground onto yet another novel background.

Splits and labels. We retain the original class labels and test splits from UCF101 and HMDB51, ensuring that results are directly comparable to prior work.

3.3 Evaluation Protocol

We report the following metrics:

- **IID Accuracy**: Top-1 accuracy on the original (unmodified) test set, measuring standard recognition performance.
- **Avg SCUBA**: Top-1 accuracy averaged across all three background sources, measuring robustness to background distribution shift.
- **Avg SCUFO**: Top-1 accuracy averaged across background sources on static (tiled-frame) clips. Lower values indicate less reliance on static foreground cues.
- **Confl-FG**: Top-1 accuracy on the conflicting-foreground condition.
- **Contrasted Accuracy**: The fraction of test samples correctly classified on SCUBA *and* incorrectly classified on SCUFO. This metric directly quantifies the extent to which a model relies on temporal dynamics rather than static appearance, since high contrasted accuracy requires succeeding when backgrounds change (SCUBA) while failing when only static cues remain (SCUFO).

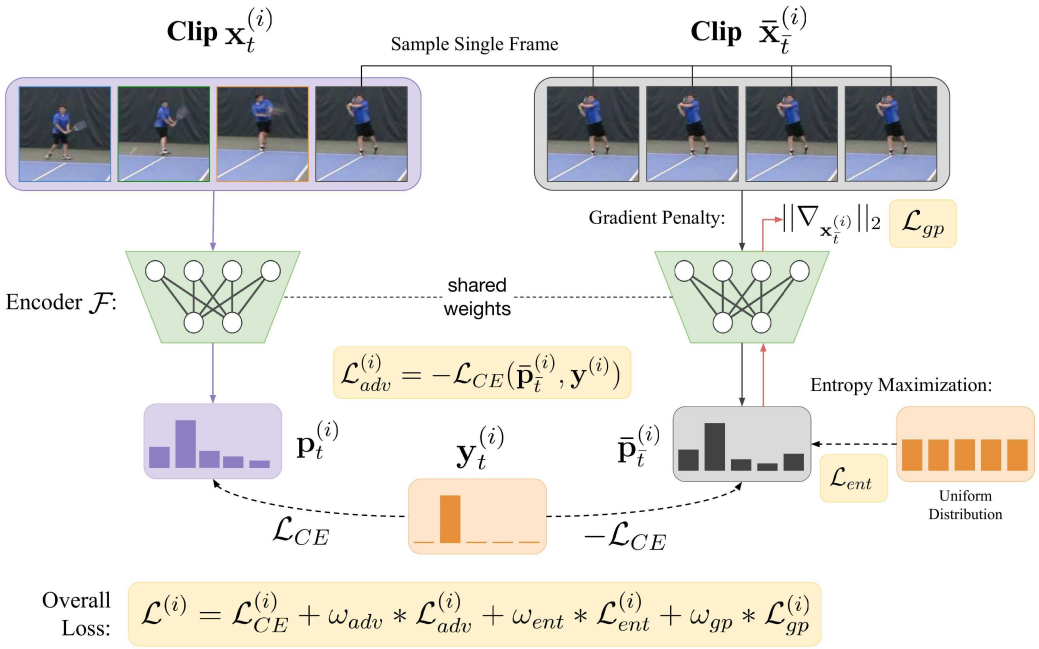


Figure 2: Overview of the ALBAR training architecture. Given video clip $\mathbf{x}_t^{(i)}$, we sample a random frame and stack it to create static clip $\bar{\mathbf{x}}_{\bar{t}}^{(i)}$. The shared encoder processes both clips, with the motion clip optimized for classification and the static clip subject to adversarial and entropy losses that suppress static-cue reliance.

Contrasted accuracy is our primary diagnostic metric: a model that exploits motion should score high on SCUBA and low on SCUFO, yielding high contrasted accuracy, whereas a model relying on static shortcuts will have low contrasted accuracy regardless of its IID performance.

4 Method

We present ALBAR (**A**dversarial **L**earning approach to mitigate **B**iases in **A**ction **R**ecognition), which suppresses static-cue shortcuts without requiring bias attribute labels, auxiliary discriminators, or architectural modifications beyond the standard video encoder. The key idea is to adversarially train the classifier so that it is uninformative on motionless inputs while remaining discriminative on motion clips.

4.1 Architecture and Static Clip Construction

We use a Video Swin Transformer (Swin3D-T) (Liu et al., 2022) pretrained on Kinetics-400 (Carreira & Zisserman, 2017) as the backbone encoder f_θ. For each training video clip $\mathbf{x}_t^{(i)} \in \mathbb{R}^{T\times H\times W\times 3}$ with label $y^{(i)}$, we construct a static counterpart $\bar{\mathbf{x}}_{\bar{t}}^{(i)}$ by selecting the middle frame and tiling it T times to match the clip length (Figure 2). The static clip preserves all spatial information—object appearance, scene layout, actor pose—but removes temporal dynamics. Both clips are processed by the same encoder f_θ, producing logits $\mathbf{z} = f_\theta(\mathbf{x}_t^{(i)})$ for the motion clip and $\bar{\mathbf{z}} = f_\theta(\bar{\mathbf{x}}_{\bar{t}}^{(i)})$ for the static clip.

We select the middle frame rather than a random or boundary frame because it typically captures the most representative pose of the action; our ablation study (Section 5) confirms that this choice performs best or comparably across benchmarks.

4.2 LOSS OBJECTIVES

ALBAR combines four loss terms that jointly encourage the encoder to be discriminative on motion clips while being maximally uninformative on static clips.

Temporal classification loss. The standard cross-entropy loss on motion clips ensures that the model learns to classify actions from temporal inputs:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} \mathbf{1}[y^{(i)} = c] \log p(y = c \mid \mathbf{x}_t^{(i)}), \tag{1}$$

where $p(y \mid \mathbf{x}_t^{(i)}) = \text{softmax}(f_\theta(\mathbf{x}_t^{(i)}))$ and C is the number of classes.

Adversarial static loss. To penalize confident predictions on static clips, we apply a negative cross-entropy loss (Xie et al., 2017):

$$\mathcal{L}_{\text{adv}} = -\sum_{c=1}^{C} \mathbf{1}[y^{(i)} = c] \log p(y = c \mid \bar{\mathbf{x}}_{\bar{t}}^{(i)}). \tag{2}$$

Unlike gradient reversal approaches (Goodfellow et al., 2014), we directly negate the cross-entropy on static clips: minimizing $-\mathcal{L}_{\text{adv}}$ pushes the encoder away from correct classification on motionless inputs. This formulation is simpler and avoids the instability of gradient reversal layers, as it requires no additional discriminator network.

Entropy maximization loss. To complement the adversarial loss, we maximize the entropy of predictions on static clips, driving the output distribution toward uniformity over classes:

$$\mathcal{L}_{\text{ent}} = \sum_{c=1}^{C} p(y = c \mid \bar{\mathbf{x}}_{\bar{t}}^{(i)}) \log p(y = c \mid \bar{\mathbf{x}}_{\bar{t}}^{(i)}). \tag{3}$$

Note that $\mathcal{L}_{\text{ent}}$ is the negative entropy of the predicted distribution; minimizing it with a positive weight in the total loss therefore maximizes entropy. While the adversarial loss discourages the correct class, the entropy term additionally prevents the model from concentrating probability mass on any incorrect class, ensuring that static clips yield truly uninformative predictions.

Gradient penalty. Adversarial training can destabilize optimization. We regularize the encoder by penalizing the magnitude of gradients of the logit output with respect to the static clip input:

$$\mathcal{L}_{\text{gp}} = \frac{1}{B} \sum_{i=1}^{B} \sqrt{\sum_{j} \left(\frac{\partial \bar{z}_j^{(i)}}{\partial \bar{\mathbf{x}}_{\bar{t}}^{(i)}} \right)^2}, \tag{4}$$

where $\bar{z}_j^{(i)}$ denotes the j-th logit for the i-th static clip and B is the batch size. This penalty encourages the encoder to produce smooth, slowly varying outputs in response to static inputs, preventing oscillatory adversarial dynamics.

4.3 TOTAL OBJECTIVE AND TRAINING

The overall training loss is:

$$\mathcal{L} = \lambda_{\text{CE}} \, \mathcal{L}_{\text{CE}} \; - \; \lambda_{\text{adv}} \, \mathcal{L}_{\text{adv}} \; + \; \lambda_{\text{ent}} \, \mathcal{L}_{\text{ent}} \; + \; \lambda_{\text{gp}} \, \mathcal{L}_{\text{gp}}, \tag{5}$$

where λ_{CE}, λ_{adv}, λ_{ent}, and λ_{gp} are scalar weights. The negative sign before $\lambda_{\text{adv}}\mathcal{L}_{\text{adv}}$ reflects that $\mathcal{L}_{\text{adv}}$ is itself the negative cross-entropy (Equation 2), so the combined effect is to *subtract* the standard cross-entropy on static clips from the total loss, penalizing correct classification on motionless inputs.

ALBAR requires no bias attribute annotations and introduces no additional network components beyond the standard encoder and classification head. It is also compatible with data augmentation strategies such as StillMix (Li et al., 2023), which can be combined to achieve further debiasing gains. The entire training procedure operates end-to-end with standard stochastic gradient descent on the joint objective in Equation 5.

Table 1: Comparison results on IID and OOD test sets of various debiasing methods on HMDB51. All experiments use Swin-T pretrained using Kinetics-400. The light blue column highlights the contrasted accuracy.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	73.92	43.93	20.46	36.58	27.84
$\text{Mixup}_{\text{ICLR'18}}$	74.58	43.10	21.17	36.62	26.09
$\text{VideoMix}_{\text{arXiv'20}}$	73.31	39.39	20.44	32.68	23.13
$\text{SDN}_{\text{NeurIPS'19}}$	74.66	40.02	20.22	34.87	22.88
$\text{BE}_{\text{CVPR'21}}$	74.31	43.56	19.96	35.99	27.84
$\text{ActorCutMix}_{\text{CVIU'23}}$	74.05	46.79	22.07	36.97	28.12
$\text{FAME}_{\text{CVPR'22}}$	73.79	51.40	26.92	39.61	29.66
$\text{StillMix}_{\text{ICCV'23}}$	**74.82**	51.81	13.39	47.38	40.28
Ours	72.81	**53.53** ↑21.9%	**1.50** ↓92.7%	**48.13** ↑31.4%	**53.22** ↑91.2%
Ours w/ StillMix aug.	74.31	54.24	1.35	47.07	53.68

5 Experiments

5.1 Experimental Setup

Model. All experiments use a Video Swin Transformer (Swin3D-T) (Liu et al., 2022) pretrained on Kinetics-400 (Carreira & Zisserman, 2017) as the backbone encoder. This architecture is used consistently across all compared methods to ensure fair comparison.

Datasets. We evaluate on HMDB51 (Kuehne et al., 2011) and UCF101 (Soomro et al., 2012). For UCF101, we use our improved segmentation-based SCUBA/SCUFO benchmark described in Section 3. For downstream task evaluation, we use the UCF-Crime dataset (Sultani et al., 2018) for weakly-supervised video anomaly detection and THUMOS14 (Jiang et al., 2014) for temporal action localization.

Baselines. We compare ALBAR against: (i) no debiasing (None), (ii) Mixup (Zhang et al., 2017), (iii) VideoMix (Yun et al., 2020), (iv) Scene Debiasing Network (SDN) (Choi et al., 2019), (v) Background Erasing (BE) (Wang et al., 2021), (vi) ActorCutMix, (vii) FAME (Ding et al., 2022), and (viii) StillMix (Li et al., 2023). All methods use the same Swin3D-T backbone.

Metrics. We report IID Top-1 accuracy, Avg SCUBA (↑), Avg SCUFO (↓), Confl-FG (↑), and Contrasted Accuracy (↑), as described in Section 3.3. Contrasted Accuracy is our primary OOD metric.

5.2 Main Results on HMDB51

Table 1 presents comparison results on HMDB51. ALBAR achieves a contrasted accuracy of **53.22%**, surpassing the previous state-of-the-art StillMix (Li et al., 2023) by 12.94 absolute percentage points—a 91.2% relative improvement.

Several key observations emerge. Augmentation-based methods (Mixup, VideoMix) do not improve contrasted accuracy substantially, with some even degrading it. Scene-adversarial methods (SDN) and background-focused approaches (BE, FAME) improve SCUBA accuracy but leave foreground static shortcuts largely intact, as evidenced by high SCUFO accuracy. StillMix is the strongest prior approach, reducing SCUFO to 13.39% and reaching 40.28% contrasted accuracy. However, ALBAR drives SCUFO down to **1.50%**—a 92.7% relative reduction—near-completely suppressing foreground static shortcuts. ALBAR also achieves the highest Avg SCUBA (53.53%) and Confl-FG (48.13%), indicating strong robustness to both background and foreground bias.

ALBAR incurs a small IID accuracy trade-off (72.81% vs. 74.82% for StillMix), but when combined with StillMix augmentation (ALBAR w/ StillMix), IID accuracy recovers to 74.31% and contrasted

Table 2: Results on IID and OOD test sets of various debiasing methods on UCF101 using our proposed updated protocol. The light blue column highlights the contrasted accuracy. All experiments use Swin-T pretrained using Kinetics-400.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	95.51	18.78	1.50	49.36	17.53
StillMix ICCV'23	**96.14**	24.68	0.36	**55.03**	24.40
Ours	94.98	**26.25** ↑39.8%	**0.14** ↓90.7%	54.02 ↑9.4%	**26.23** ↑49.6%
Ours w/ StillMix aug.	94.71	31.17	0.16	58.84	31.14

accuracy further improves to 53.68%, confirming that ALBAR is compatible with augmentation-based methods and can benefit from them.

5.3 Results on UCF101 with Improved Benchmark

Table 2 shows results on our improved UCF101 benchmark. The improved segmentation-based protocol provides a leakage-free evaluation surface.

ALBAR achieves a contrasted accuracy of **26.23%**, compared to 17.53% for the undebiased baseline (+49.6% relative) and 24.40% for StillMix. Our method achieves the best Avg SCUBA (26.25%) and the lowest Avg SCUFO (0.14%), while maintaining 94.98% IID accuracy (vs. 95.51% baseline, a negligible 0.53 point drop). Combined with StillMix augmentation, the contrasted accuracy improves further to 31.14%, with IID accuracy of 94.71%. The consistent gains across both datasets confirm that ALBAR's debiasing effect is robust and general.

5.4 Ablation Studies

5.4.1 Loss Component Analysis

Table 3 ablates the contribution of each loss component in ALBAR on HMDB51.

Table 3: Ablation to determine the effectiveness of each component loss objective.

	$\mathcal{L}_{CE}$	$\mathcal{L}_{adv}$	$\mathcal{L}_{ent}$	$\mathcal{L}_{gp}$	IID	OOD			
						Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
(a)	✓	✗	✗	✗	73.92	43.93	20.46	36.58	27.84
(b)	✓	✓	✗	✗	72.61	42.29	**0.00**	32.77	42.29
(c)	✓	✗	✓	✗	71.57	46.02	2.97	38.40	44.30
(d)	✓	✗	✗	✓	72.09	41.13	14.69	35.16	29.24
(e)	✓	✓	✓	✗	72.22	46.81	0.49	41.13	46.72
(f)	✓	✓	✗	✓	71.70	45.92	**0.00**	38.52	45.92
(g)	✓	✗	✓	✓	**73.86**	49.14	9.74	42.89	41.11
(h)	✓	✓	✓	✓	72.81	**53.53**	1.50	**48.13**	**53.22**

Starting from the cross-entropy-only baseline (row a: 27.84% contrasted accuracy), each adversarial component provides meaningful improvements. The adversarial loss alone (row b) achieves 42.29% contrasted accuracy and drives SCUFO to 0.00%, demonstrating strong static-cue suppression. The entropy maximization loss alone (row c) achieves 44.30% contrasted accuracy, with complementary improvement in SCUBA accuracy. The gradient penalty alone (row d) yields only a modest improvement (29.24%), consistent with its role as a stabilizer rather than a primary debiasing signal.

Pairwise combinations further improve results: $\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{ent}}$ (row e) reaches 46.72% contrasted accuracy with very low SCUFO (0.49%), while $\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{gp}}$ (row f) achieves 45.92%. The full ALBAR model (row h), combining all four losses, achieves the best contrasted accuracy (53.22%) and Avg SCUBA (53.53%). The gradient penalty plays a critical role in the full model: comparing rows (e) and (h) shows that adding $\mathcal{L}_{\text{gp}}$ significantly boosts SCUBA accuracy while keeping SCUFO low, suggesting it helps stabilize training and improves the quality of motion-grounded features.

5.4.2 Static Frame Sampling Strategy

Table 4 evaluates the choice of frame used to construct the static clip.

Table 4: Ablation to evaluate static frame sampling strategy.

Frame Sampling Strategy	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
Random	73.20	53.22	0.42	49.84	53.02
First	72.75	50.92	0.18	45.59	50.91
Middle	72.81	53.53	1.50	48.13	**53.22**
Last	72.68	49.49	0.40	42.42	49.40

We compare random frame selection, the first frame, the middle frame, and the last frame. The middle frame strategy achieves the best contrasted accuracy (53.22%), closely followed by random selection (53.02%). First and last frame strategies perform slightly worse, likely because the middle frame captures a more representative pose of the actor during the peak of the action, while boundary frames may capture less informative pre- or post-action states. We adopt the middle frame as our default.

5.5 Downstream Task Transfer

Table 5 evaluates whether ALBAR's debiased representations transfer to downstream video understanding tasks.

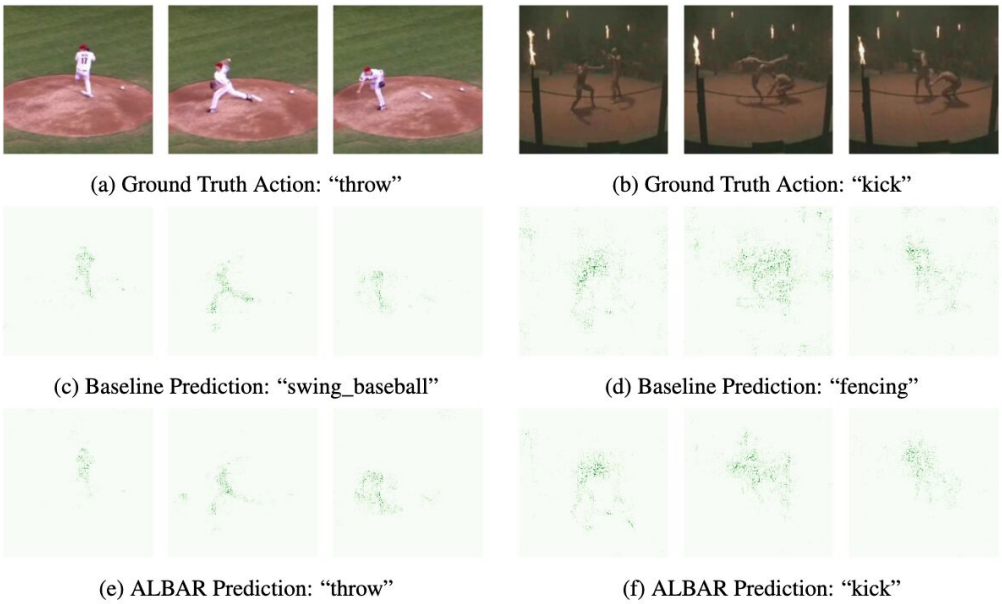
Table 5: Evaluation of our debiased model across downstream anomaly detection (UCF_Crime) and temporal action localization (THUMOS14) tasks.

Method	**Action Recognition HMDB51 (OOD)**	**Anomaly Detection UCF_Crime**	**Temporal Action Localization THUMOS14**
Baseline	27.84	82.39	54.89
Ours	**53.22**	**84.91**	**55.20**

On UCF-Crime anomaly detection (Sultani et al., 2018), ALBAR improves the AUC from 82.39 to **84.91**. On THUMOS14 temporal action localization (Jiang et al., 2014), the mAP improves from 54.89 to **55.20**. These consistent improvements confirm that learning to rely on motion-grounded features—rather than static appearance—yields representations that are more sensitive to the temporal dynamics underlying anomalous events and action boundaries, which are both inherently temporal phenomena. The fact that debiasing on HMDB51 improves performance on these separate tasks underscores the generality of motion-grounded representations.

5.6 Qualitative Analysis

Figure 3 illustrates failure cases of the baseline model that ALBAR corrects. The baseline model systematically assigns action labels correlated with background or foreground appearance, even when the depicted motion is inconsistent with that label. ALBAR, having been trained to suppress the static-cue pathway, correctly identifies the action from motion patterns alone. These qualitative examples are consistent with the quantitative findings: the near-zero SCUFO accuracy of ALBAR indicates that the model extracts almost no discriminative information from motionless inputs, which translates to more accurate and contextually robust predictions in real videos.



(a) Ground Truth Action: “throw”

(b) Ground Truth Action: “kick”

(c) Baseline Prediction: “swing_baseball”

(d) Baseline Prediction: “fencing”

(e) ALBAR Prediction: “throw”

(f) ALBAR Prediction: “kick”

Figure 3: Qualitative examples from the HMDB51 test set. The baseline model selects incorrect action labels due to static spatial context (e.g., assigning a background-correlated label when context cues are misleading), while ALBAR correctly identifies the action by attending to motion features.

6 Conclusion

We introduced ALBAR, an adversarial learning framework for debiasing action recognition models against both background and foreground static shortcuts. By constructing motionless counterpart clips for each training video and penalizing confident predictions on these static inputs through negative cross-entropy, entropy maximization, and a gradient penalty, ALBAR forces the model to ground its predictions in temporal dynamics. The method requires no bias attribute annotations, no auxiliary discriminators, and is compatible with augmentation-based debiasing methods.

Alongside ALBAR, we introduced an improved UCF101-SCUBA/SCUFO benchmark that replaces coarse bounding-box foreground masks with fine-grained per-frame actor segmentations from SAM-Track (Cheng et al., 2023), eliminating the background leakage that has compromised prior evaluations. Our benchmark employs diverse backgrounds (Places365 (Zhou et al., 2017), VQGAN-CLIP (Crowson et al., 2022), sinusoidal textures) and a contrasted accuracy metric that jointly measures background robustness and foreground-bias suppression.

Experimentally, ALBAR achieves state-of-the-art results on HMDB51 (53.22% contrasted accuracy, +12.94 points over StillMix) and UCF101 (26.23% contrasted accuracy, +8.70 points over the baseline), with minimal IID degradation. Debiased representations improve downstream anomaly detection and temporal localization performance. Together, ALBAR and the improved benchmark constitute a step toward motion-grounded, contextually robust action recognition—one where models succeed because they understand what is happening, not merely where it is happening.

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